

DM583 DATA MINING

Exam Cheat Sheet & Formula Reference

1. EM-GMM CLUSTERING

- **Bayes' Rule:** Used in the *E-step* to calculate posterior probabilities[cite: 1]:

$$P(C_i|x) = [\pi_i * f(x|C_i)] / \sum [\pi_j * f(x|C_j)]$$
- **Covariance & Independence:** GMM does *not* assume variable independence[cite: 1]. Non-diagonal matrices capture correlations between dimensions[cite: 1].
- **Diagonal Covariance:** Produces **axis-aligned elliptical** clusters, not necessarily spherical ones (unless variances are equal across axes)[cite: 1].
- **Initialization:** Highly sensitive to initial parameter seeds (like k-Means)[cite: 1].

2. UNIVARIATE GAUSSIAN MLE

- Maximum Likelihood Estimators for a 1D Normal distribution are strictly the sample mean and sample variance[cite: 1]:

$$\mu = (1/N) * \sum x_i$$

$$\sigma^2 = (1/N) * \sum (x_i - \mu)^2$$
- Uses population denominator N, not sample correction N-1[cite: 1].

3. HIERARCHICAL LINKAGE (1D)

- **Single-Linkage:** Distance = minimum distance between any pair of points ($d_{\min}$)[cite: 1].
- **Complete-Linkage:** Distance = maximum distance between any pair of points ($d_{\max}$)[cite: 1].
- **1D Property:** Being 1D does **not** guarantee identical results[cite: 1]. As clusters grow, $d_{\min}$ and $d_{\max}$ diverge.

4. PARALLEL K-MEANS

- Nodes compute local sums & counts; Reducer aggregates them[cite: 1]:

$$\text{New Centroid} = (\sum \text{Partial Sums}) / (\sum \text{Partial Counts})$$
- Data splits do **not** need to be equal[cite: 1]. Odd sample sizes (e.g., N=7) are handled without issue[cite: 1].

5. KNN OUTLIERS & MINKOWSKI

- **KNN Outlier Score:** Distance to the k -th nearest neighbor[cite: 1].
- **Minkowski Distance Formula:**

$$D(x, y) = [\sum |x_i - y_i|^p]^{(1/p)}$$
- **1D Simplification:** In 1D, for all $p \geq 1$, it reduces to absolute difference: $|x - y|$ [cite: 1]. Choice of p has zero effect in 1D[cite: 1].

6. DBSCAN DEFINITIONS

- **ϵ (Epsilon):** Search radius around a point[cite: 1].
- **MinPts:** Minimum points within ϵ -neighborhood for a core point[cite: 1].
- **Core Point:** Neighbors $\geq$ MinPts[cite: 1].
- **Border Point:** Neighbors $<$ MinPts, but within ϵ of a Core Point[cite: 1].
- **Noise Point:** Neither core nor border[cite: 1].
- Increasing MinPts turns isolated points into noise instead of single-point clusters[cite: 1].

7. APRIORI & ASSOCIATION MINING

- **Apriori Principle:** All subsets of a frequent itemset must be frequent[cite: 1]. If any subset is infrequent, the candidate is pruned *before* database lookup[cite: 1].
- **Maximal Frequent:** Frequent itemset with **no frequent supersets**[cite: 1].
- **Closed Frequent:** Frequent itemset where **no superset has the exact same support count**[cite: 1].
- **Rule Metrics ($A \Rightarrow B$):**

$$\text{Support} = \text{Count}(A \cup B)$$

$$\text{Confidence} = \text{Count}(A \cup B) / \text{Count}(A)$$

TOPIC	QUESTION STATEMENT SUMMARY	VERDICT	CORE REASON
Q1.1 EM-GMM	EM-GMM makes use of Bayes' rule.[cite: 1]	TRUE	Used in the E-step for posterior probabilities.[cite: 1]
Q1.2 EM-GMM	Assumes zero covariance within clusters.[cite: 1]	FALSE	GMMs allow non-diagonal covariance matrices.[cite: 1]
Q1.3 EM-GMM	Given priors & densities, posteriors ≈ 0.213 and 0.787 .[cite: 1]	TRUE	$0.3(0.0086) / [0.3(0.0086) + 0.7(0.0136)] = 0.213$ [cite: 1]
Q1.4 EM-GMM	Diagonal covariance guarantees spherical clusters.[cite: 1]	FALSE	Yields axis-aligned elliptical clusters if variances differ.[cite: 1]
Q1.5 EM-GMM	EM-GMM is not sensitive to initial cluster centers.[cite: 1]	FALSE	Highly sensitive to initialization (like k-Means).[cite: 1]
Q2.1 Gaussian MLE	1D Normal MLE uses sample mean and sample variance.[cite: 1]	TRUE	Matches exact Gaussian MLE definitions.[cite: 1]
Q2.2 Hierarchical	1D data guarantees identical Single & Complete linkage.[cite: 1]	FALSE	Cluster distance definitions diverge as clusters merge.[cite: 1]
Q2.3 Parallel k-Means	Odd sample size (N=7) prevents distributed execution.[cite: 1]	FALSE	Nodes process unequal partition sizes without issue.[cite: 1]

TOPIC	QUESTION STATEMENT SUMMARY	VERDICT	CORE REASON
Q2.4 KNN Outliers	$k=1$ scores in 1D dataset equal {10, 10, 20, 15, 15, 5, 5}.[cite: 1]	TRUE	Minkowski reduces to absolute distance $ x - y $ in 1D.[cite: 1]
Q2.5 DBSCAN	MinPts=1 and MinPts=2 both yield 4 identical clusters.[cite: 1]	FALSE	MinPts=2 turns point {10} into noise (yields 3 clusters).[cite: 1]
Q3.1 Apriori	Computes support for {BUTTER, DIAPERS, MILK}.[cite: 1]	FALSE	Pruned beforehand because subset {BUTTER, MILK} is infrequent.[cite: 1]
Q3.2 Apriori	Generates and tests {BREAD, BUTTER, MILK}.[cite: 1]	FALSE	Pruned beforehand because subset {BUTTER, MILK} is infrequent.[cite: 1]
Q3.3 Maximal	{DIAPERS, MILK} is proven maximal without DB lookup.[cite: 1]	FALSE	{BREAD, DIAPERS, MILK} could still turn out to be frequent.[cite: 1]
Q3.4 Closed	{BREAD, DIAPERS} is a closed frequent itemset.[cite: 1]	TRUE	No superset retains its exact support count of 6.[cite: 1]
Q3.5 Association	{DIAPERS} $\Rightarrow$ {BREAD, MILK} has Support=5, Confidence=5/9.[cite: 1]	TRUE	Full set support = 5; Antecedent support = 9.[cite: 1]